



UNIT SAINS KOMPUTER  
KOLEJ MATRIKULASI MELAKA  
SESI 2024/2025

ANSWER SCHEME  
DC014 (IQUA)

| Question No. | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mark                     |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1 (a)        | Animation<br>Video<br>Audio<br>Graphic<br><br><i>*Text no acceptable</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | J1<br>Choose any one (1) |
| 1 (b)        | Animation<br>- Animation is a process of making a static image look like it is moving.<br>- Animation is a series of graphical images displayed one after another to simulate movement.<br>- Animation is useful for illustrating concepts which involve movement.<br>- Can help organize thoughts and illustrate processes.<br>- To make a presentation livelier and present complex concept briefly.<br><br>Video<br>- Video show the real situation of the event hence help the viewers retain more of what they see.<br>- Video can deliver messages more accurate within a short time.<br>- Video support to reinforce the story.<br>- Provide realistic impact to the simulation or presentation<br>- Bring the user closer to real world.<br><br>Audio<br>- helps to convey information more effectively.<br>- able to increase motivation and make people more interested in following the delivery of information.<br>- audio can provide a visual representation more effectively than reading alone.<br><br>Graphic<br>- Used to convey messages as it serves the purpose of presenting information more attractively & effectively.<br>- To make multimedia applications more attractive.<br>- To illustrate ideas through still pictures.<br>- To convey information faster and easier. | J2<br>Choose any one (1) |
| 1 (c) (i)    | length, width, height, and time/motion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | J2                       |
| 1 (c) (ii)   | 3D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | J1                       |
| 2 (a) (i)    | Phase : Production<br>Activity : Recording Audio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | J2                       |



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| Question No.                                                                                                        | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mark            |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------|
| 2 (a) (ii)                                                                                                          | Phase : Post- production<br>Activity : Distribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | J2              |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| 2 (a) (iii)                                                                                                         | Phase : Pre-production<br>Activity : Planning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | J2              |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| 2 (b) (i)                                                                                                           | <b>FIGURE 1(a):</b> Time based authoring tool<br><b>FIGURE 1(b):</b> Page-based authoring tool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | J1<br>J1        |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| 2 (b) (ii)                                                                                                          | <b>FIGURE 1(a):</b> The multimedia elements and events are presented and organized along a timeline.<br><b>FIGURE 1(b):</b> The multimedia elements and events are presented and organized as pages of book or stack of cards.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | J1<br>J1        |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| 2 (c)                                                                                                               | <b>Z:</b> Audio navigation refers to control and handling multimedia applications using audio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | J2              |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| 2 (d)                                                                                                               | <b>Capacity:</b> Can store high end multimedia elements such as video<br><b>Speed data of transfer rate:</b> Relatively faster data transfer rate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | J2<br>J2        |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| 3 (a)                                                                                                               | <table><tr><th>File Processing</th><th>Database System</th></tr><tr><td>Each department has its own set of data files.<br/>✓ limited data sharing</td><td>Data can be shared between authorized users of the database. All the users have their own right to access the database</td></tr><tr><td>Redundant data<br/>✓ same fields are stored in multiple files</td><td>Reduce data redundancy<br/>✓ most of data are stored in only one file.</td></tr><tr><td>Lack of data integrity<br/>✓ user make charges data that are stored in multiple file that increase chance of errors.</td><td>Improve data integrity<br/>✓ user make changes to one file instead of multiple file</td></tr><tr><td>Poor data access<br/>✓ difficult to access data that is stored in separate files in different department.</td><td>Easier data access<br/>✓ data stored in database are shared over network easily access by user.</td></tr></table> | File Processing | Database System | Each department has its own set of data files.<br>✓ limited data sharing | Data can be shared between authorized users of the database. All the users have their own right to access the database | Redundant data<br>✓ same fields are stored in multiple files | Reduce data redundancy<br>✓ most of data are stored in only one file. | Lack of data integrity<br>✓ user make charges data that are stored in multiple file that increase chance of errors. | Improve data integrity<br>✓ user make changes to one file instead of multiple file | Poor data access<br>✓ difficult to access data that is stored in separate files in different department. | Easier data access<br>✓ data stored in database are shared over network easily access by user. | J2,J2<br>Choose any one (1) |
| File Processing                                                                                                     | Database System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| Each department has its own set of data files.<br>✓ limited data sharing                                            | Data can be shared between authorized users of the database. All the users have their own right to access the database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| Redundant data<br>✓ same fields are stored in multiple files                                                        | Reduce data redundancy<br>✓ most of data are stored in only one file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| Lack of data integrity<br>✓ user make charges data that are stored in multiple file that increase chance of errors. | Improve data integrity<br>✓ user make changes to one file instead of multiple file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |
| Poor data access<br>✓ difficult to access data that is stored in separate files in different department.            | Easier data access<br>✓ data stored in database are shared over network easily access by user.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                 |                                                                          |                                                                                                                        |                                                              |                                                                       |                                                                                                                     |                                                                                    |                                                                                                          |                                                                                                |                             |



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| Question No.                                                                                               | Answer                                                                                                                                                                                                                                                                                                                                                                                                           | Mark                           |                    |                                                                                                            |                                                                                          |        |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|
| 3 (b)                                                                                                      | <ul style="list-style-type: none"><li>Database systems are complex, difficult, and time consuming to design.</li><li>Substantial hardware and software start-up cost</li><li>Damage to database affects virtually all applications program</li><li>Extensive conversion costs in moving from a file-based system to a database system.</li><li>Initial training required for all programmers and users</li></ul> | J1<br>J1<br>Choose any two (2) |                    |                                                                                                            |                                                                                          |        |
| 4 (a)                                                                                                      | <ul style="list-style-type: none"><li>Data integrity refers to the correctness and consistency of stored data.</li><li>Data integrity refers to the accuracy, consistency, and reliability of data.</li></ul>                                                                                                                                                                                                    | J2<br>Choose any one (1)       |                    |                                                                                                            |                                                                                          |        |
| 4 (b)(i)                                                                                                   | Query                                                                                                                                                                                                                                                                                                                                                                                                            | J1                             |                    |                                                                                                            |                                                                                          |        |
| 4 (b)(ii)                                                                                                  | Form                                                                                                                                                                                                                                                                                                                                                                                                             | J1                             |                    |                                                                                                            |                                                                                          |        |
| 5 (a)(i)                                                                                                   | GPS is a satellite navigation system used to determine the ground position of an object.                                                                                                                                                                                                                                                                                                                         | J2                             |                    |                                                                                                            |                                                                                          |        |
| 5 (a)(ii)                                                                                                  | Television broadcasting is a telecommunications network for distribution of television program content, where a central operation provides programming to many television stations providers.                                                                                                                                                                                                                    | J2                             |                    |                                                                                                            |                                                                                          |        |
| 5 (b)(i)                                                                                                   | X: Wireless networking media<br>Y: Wired networking media                                                                                                                                                                                                                                                                                                                                                        | J1<br>J1                       |                    |                                                                                                            |                                                                                          |        |
| 5 (b)(ii)                                                                                                  | <table><tr><th>Networking Media X</th><th>Networking Media Y</th></tr><tr><td>Wireless transmission media allows devices to communicate with each other without any physical connection.</td><td>Wired transmission media requires a physical connection (cable/bounded) between devices.</td></tr></table>                                                                                                      | Networking Media X             | Networking Media Y | Wireless transmission media allows devices to communicate with each other without any physical connection. | Wired transmission media requires a physical connection (cable/bounded) between devices. | J2, J2 |
| Networking Media X                                                                                         | Networking Media Y                                                                                                                                                                                                                                                                                                                                                                                               |                                |                    |                                                                                                            |                                                                                          |        |
| Wireless transmission media allows devices to communicate with each other without any physical connection. | Wired transmission media requires a physical connection (cable/bounded) between devices.                                                                                                                                                                                                                                                                                                                         |                                |                    |                                                                                                            |                                                                                          |        |
| 5 (b)(iii)                                                                                                 | <ul style="list-style-type: none"><li>Cellular Radio Transmission</li><li>Microwave &amp; satellite Transmission</li><li>Infrared Transmission</li></ul>                                                                                                                                                                                                                                                         | J1<br>J1<br>Choose any two (2) |                    |                                                                                                            |                                                                                          |        |
| 5 (c)(i)                                                                                                   | AA: Modem<br>BB: Router / Wireless Router                                                                                                                                                                                                                                                                                                                                                                        | J1<br>J1                       |                    |                                                                                                            |                                                                                          |        |
| 5 (c)(ii)                                                                                                  | Type: Wireless<br>Example: Cellular Radio Transmission                                                                                                                                                                                                                                                                                                                                                           | J1<br>J1                       |                    |                                                                                                            |                                                                                          |        |
| 5 (d)(i)                                                                                                   | Repeater                                                                                                                                                                                                                                                                                                                                                                                                         | J1                             |                    |                                                                                                            |                                                                                          |        |
| 5 (d)(ii)                                                                                                  | Wireless Access Point                                                                                                                                                                                                                                                                                                                                                                                            | J1                             |                    |                                                                                                            |                                                                                          |        |
| 6 (a)(i)                                                                                                   | Search engine                                                                                                                                                                                                                                                                                                                                                                                                    | J1                             |                    |                                                                                                            |                                                                                          |        |
| 6 (a)(ii)                                                                                                  | A search tool that uses keywords to find information that is related to a specific topic.                                                                                                                                                                                                                                                                                                                        | J2                             |                    |                                                                                                            |                                                                                          |        |



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|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 6 (a)(iii)   | <ul style="list-style-type: none"> <li>Google search</li> <li>Yahoo! Search</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | J1<br>Choose anyone (1)        |
| 6 (b)        | <ul style="list-style-type: none"> <li>Easy Access: Search engines let you quickly find information online without needing to know specific websites.</li> <li>Fast Results: Just type in what you're looking for, and within seconds, you'll see a list of relevant websites.</li> <li>Best Picks First: Search engines show you the most useful websites at the top of the list, saving you time from scrolling through less helpful ones.</li> <li>Refine Your Search: You can easily filter results by date, location, or type of content to find exactly what you need.</li> <li>Always Fresh: Search engines keep updating to show you the latest news and information, so you're always in the know.</li> </ul> <p>OR any acceptable answer.</p> | J2<br>Choose anyone (1)        |
| 6 (c)(i)     | Using phrases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | J1                             |
| 6 (c)(ii)    | Using Boolean operator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | J1                             |
| 6 (c)(iii)   | Using field                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | J1                             |
| 6 (d)(i)     | <ul style="list-style-type: none"> <li>Online Communication</li> <li>Application for communication</li> <li>Online forum</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | J1<br>Choose any one (1)       |
| 6 (d)(ii)    | Online Entertainment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | J1                             |
| 6 (d)(iii)   | e-Commerce                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | J1                             |
| 6 (d)(iv)    | <ul style="list-style-type: none"> <li>Online Education &amp; Writing</li> <li>Online Education</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | J1<br>Choose any one (1)       |
| 6 (d)(v)     | <ul style="list-style-type: none"> <li>Social Media</li> <li>Social Networking</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | J1<br>Choose any one (1)       |
| 6 (e)(i)     | <ul style="list-style-type: none"> <li>Regularly clear browsing history and cookies.</li> <li>Use a trusted web browser</li> <li>Use private browsing mode such as Incognito for Google</li> </ul> <p>OR any acceptable answer</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | J1<br>J1<br>Choose any two (2) |
| 6 (e)(ii)    | E-mail privacy refers to the protection of confidentiality, integrity, and security of electronic mail communications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | J2                             |



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|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 6 (f)        | <ul style="list-style-type: none"><li>• Remote Monitoring: Devices let doctors check on patients' health from a far.</li><li>• Personalized Care: Devices give doctors personalized info to tailor treatment plans.</li><li>• Better Outcomes: Monitoring patients helps catch issues early and improve treatment results.</li><li>• Efficient Use of Resources: IoT helps hospitals manage equipment and supplies better, saving money.</li><li>• Increased Access: IoT lets people in remote areas get medical care through telemedicine.</li></ul> OR any acceptable answer | J2<br>Choose any one (1) |

**END OF ANSWER SCHEME**